PAPER_697: NGC 2525 with SN 2018gv: Type Ia Supernova Light Curve and UQFF

Class: `NGC2525WithSupernovaeSN2018gv`
CP4 Entry: #281
Keywords: NGC 2525, SN 2018gv, Type Ia supernova, Arnett rule, Phillips relation, light curve, UQFF
Session: 174 | **Version:** v5.31
Source: grok_share_ba508f76c8e.txt

Abstract

NGC 2525 is a barred spiral galaxy (SBc) at 60 Mly hosting Type Ia SN 2018gv, observed by Hubble in time-lapse. The Arnett light curve model and Phillips relation are applied with UQFF modifications to the peak luminosity.

1. Parameters

Parameter	Value
M_{galaxy}	$3 \times 10^{10} M_{\odot}$
Distance	60 Mly
$L_{SN,peak}$	1.5×10^{43} W
M_{Ni56}	$0.6 M_{\odot}$
Rise time	14 days

2. Type Ia Light Curve (Arnett)

$$L(t) \approx L_{peak}\, e^{-\frac{1}{2}\left(\frac{t-t_r}{\sigma_t}\right)^2} \times e^{-(t-t_r)/t_d}$$

3. Phillips Relation

$$M_B \approx -19.3 + 0.74(\Delta m_{15} - 1.1)$$

Δm_{15}: magnitude decline in first 15 days post-peak.

4. UQFF Luminosity Correction

$$L_{UQFF}(t) = L_{SN}(t)\left(1 + \frac{\rho_{SCm}}{\rho_{UA}}\right)(1 - f_{TRZ})$$

5. Barred Disk Rotation

$$v_c(R) = \sqrt{v_{disk}^2 + v_{bar}^2}$$

where $v_{bar}^2 = GM_{gal} \cdot 0.3(1 - e^{-R/R_{bar}})/R$.

References

- Arnett (1982), ApJ, 253, 785
- Phillips (1993), ApJ, 413, L105
- UQFF Framework, Star-Magic Session 174

Whitepaper auto-generated by _gen_whitepapers_688_701.py — Star-Magic Session 174